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Exstrophy-Epispadias Complex

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Keywords

bladder exstrophy; cloacal exstrophy; epispadias

Questions

1. What is the live birth incidence of classic bladder exstrophy?
    a. 1 in 1000
    b. 1 in 10,000
    c. 1 in 50,000
    d. 1 in 70,000
    e. 1 in 100,000
2. What is the live birth risk of bladder exstrophy in the offspring of individuals with bladder exstrophy and epispadias?
    a. 1 in 70
    b. 1 in 300
    c. 1 in 500
    d. 1 in 700
    e. 1 in 1000
3. The main theory of embryologic maldevelopment in exstrophy is that of:
    a. underdevelopment of the cloacal membrane, preventing medial migration of the mesoderm tissue and proper lower abdominal wall development.

b. overdevelopment of the cloacal membrane, preventing medial migration of the mesodermal tissue and proper lower abdominal wall development.
c. infiltration of ectoderm into the cloacal membrane.
d. infiltration of mesoderm into the cloacal membrane.
e. invasion of endoderm into the cloacal membrane.

4. In evaluating the skeletal defects of bladder exstrophy, Sponseller and colleagues (1995) [a] found that with classic bladder exstrophy, there are changes in the orientation of the pelvic bones. These include:
   a. external rotation of the posterior aspect of the pelvis of 12 degrees on each side.
   b. retroversion of the acetabulum.
   c. an 18-degree rotation of the anterior pelvis.
   d. a 30% shortening of the pubic rami in addition to a significant pubic symphyseal diastasis.
   e. all of the above.
5. Which of the following statements is TRUE regarding hernias in children with exstrophy?
   a. Identification at the time of initial closure is not possible.
   b. They are usually unilateral.
   c. They are noted in 80% of boys and 10% of girls.
   d. The orientation of the pelvic bones makes them infrequent.
   e. A patent processus vaginalis is rarely noted.
6. Which of the following statements is TRUE regarding the male genital defect in exstrophy?
   a. The posterior length of the corporeal bodies was 30% shorter than that in healthy controls.
   b. The diameter of the posterior corporeal segments was less than that in healthy controls.
   c. The shortening of the penis was due entirely to the pubic diastasis.
   d. The anterior corporeal segments are 50% shorter than those of healthy control participants.

e. The angle between the corpora cavernosa is markedly reduced in boys with exstrophy.

7. Which of the following statements best describes findings regarding the prostate in exstrophy?
    a. Volume weight and the cross-sectional area appeared healthy compared with published results from control subjects.
    b. The prostate extended circumferentially around the urethra in all patients with exstrophy.
    c. Free prostate-specific antigen (PSA) values were greater than in healthy controls, indicating recurrent injury from infection.
    d. The vas deferens and seminal vesicles were abnormal due to the effect of the exstrophic bladder.
    e. Total PSA values were not measurable in men with exstrophy.
8. Which of the following accurately describes the vagina in the female patient with bladder exstrophy?
    a. It is shorter than normal and of smaller caliber.
    b. The vaginal orifice is displaced posteriorly because of the anterior exstrophic bladder.
    c. It is shorter than normal but of normal caliber.
    d. It is longer than normal and of wider caliber.
    e. The cervix enters the posterior vaginal wall.
9. Findings regarding the structure and innervation of the exstrophic bladder included the following:
    a. The density and binding affinity of the muscarinic receptors were similar to norms.
    b. There was a decreased ratio of collagen to muscle in the exstrophic bladder.
    c. There were more myelinated nerve profiles, indicating a later developmental stage.
    d. A threefold increase in the amount of type I collagen was noted.
    e. Study of vasoactive intestinal polypeptide, protein gene product 9.5, and calcitonin gene–related peptide indicated the presence of dysinnervation.
10. Which of the following statements best describes bladder function in patients with bladder exstrophy?

a. After reconstruction, normal cystometrograms are noted in 10% to 25% of patients who are continent.
b. Before bladder neck reconstruction, 80% of patients had compliant and stable bladders.
c. Involuntary contractions were rarely noted after bladder neck reconstruction.
d. After bladder neck reconstruction, 90% of patients maintained normal bladder compliance.
e. After successful closure, ultrastructure remains abnormal in the majority.

11. The characteristic prenatal appearance of bladder exstrophy includes which of the following?
    a. Absence of bladder filling
    b. Low-set umbilicus
    c. Widening of the pubic ramus
    d. Diminutive genitalia
    e. All of the above
12. Newborn patient selection for immediate reconstruction is based on:
    a. examination of the bladder in the nursery without anesthesia.
    b. complete lack of any surface defects on examination.
    c. indentation of the bladder under anesthesia or outward bulging when the child cries.
    d. size of the phallus at birth.
    e. extent of the pubic diastasis.
13. Fundamental steps in the modern staged reconstruction of bladder exstrophy include all of the following EXCEPT
    a. Early closure of the bladder, posterior urethra, and abdominal wall
    b. Early epispadias repair at around age 1 year
    c. Conversion of the bladder exstrophy to complete epispadias
    d. Bladder neck reconstruction before the epispadias repair to provide early continence
    e. Ureteral reimplantation at the time of bladder neck reconstruction

14. What is the best treatment option at the time of birth in a child whose bladder template is judged to be too small to undergo closure?
    a. Excision of the bladder with a nonrefluxing colon conduit
    b. Immediate closure with epispadias repair to provide resistance and allow the bladder to grow
    c. Delaying closure by 4 to 6 months with reassessment to see if the bladder will grow
    d. Bladder closure, augmentation, ureteral reimplantation, and a continence procedure
    e. An osteotomy to improve the potential for successful closure
15. Combined osteotomy was developed for all of the following reasons EXCEPT
    a. This approach allows placement of an external fixator device.
    b. Superior cosmesis is provided by this approach.
    c. There is no need to turn the patient to perform an osteotomy.
    d. Pubic approximation is easier.
    e. There is less risk of malunion of the iliac wing and also less blood loss.
16. Complications associated with osteotomy and immobilization techniques include all of the following EXCEPT
    a. Skin ulceration is associated with use of mummy wrapping.
    b. Failure of the bladder and abdominal wall closure is associated with the use of spica casting.
    c. High rates of failure of reconstruction are associated with the use of osteotomy and external fixation.
    d. Transient femoral nerve palsy is associated with the use of osteotomy.
    e. Delayed union of the iliac wings may occur after the use of posterior osteotomy.
17. Which of the following statements is TRUE regarding the various described approaches to bladder exstrophy reconstruction?
    a. The Warsaw approach includes bladder neck reconstruction at the time of initial bladder closure.
    b. The Erlangen approach includes all of the features of reconstruction of the exstrophy in a single procedure.

c. The Seattle approach (Complete Primary Repair of Exstrophy [CPRE]) includes bladder neck reconstruction as part of the complete reconstruction of exstrophy.
d. Combined bladder closure and epispadias repair is performed in cases of primary exstrophy repair at birth.
e. The Warsaw approach uses the Young repair as the preferred method for epispadias reconstruction.

18. After initial primary bladder closure in the newborn, what should be done if there are recurrent urinary tract infections?
    a. Voiding cystourethrogram
    b. Computed tomography (CT) scan of the bladder
    c. Ureteral reimplantation
    d. Prophylaxis modified
    e. Cystoscopy
19. After successful bladder closure, management should include all the following EXCEPT
    a. Calibration of the urethral outlet 4 weeks after closure to ensure free drainage
    b. Ultrasound evaluation of the kidneys and bladder
    c. Intermittent antibiotics for urinary tract infections
    d. Complete bladder drainage by suprapubic tube clamping
    e. Yearly cystoscopic evaluation
20. In a patient with bladder exstrophy who undergoes more than one closure of the bladder and urethral defect, what is the chance of having adequate bladder capacity for later bladder neck reconstruction?
    a. 10%
    b. 20%
    c. 30%
    d. 60%
    e. 75%
21. The key concepts in the reconstruction of epispadias include all of the following EXCEPT
    a. Correction of ventral chordee
    b. Urethral reconstruction
    c. Glans reconstruction

d. Penile skin coverage
e. Penile lengthening

22. Information gleaned from most major series of bladder neck reconstruction indicates that the most important factor to predict success and eventual continence after bladder neck reconstruction is:
    a. the age of the child.
    b. the number of prior bladder infections.
    c. the number of attempts at bladder closure.
    d. bladder capacity.
    e. vesicoureteral reflux.
23. After bladder neck reconstruction, within what time period do the majority of patients achieve daytime continence?
    a. 2 months
    b. 6 months
    c. 1 year
    d. 2 years
    e. 4 years
24. After a failed attempted bladder closure in the newborn period, what time period should elapse before a secondary repair is attempted?
    a. 2 months
    b. 6 months
    c. 12 months
    d. 18 months
    e. 2 years
25. All of the following statements are TRUE regarding the results of modern staged reconstruction of exstrophy EXCEPT
    a. The onset of eventual continence and continence rates were unchanged in those who had initial successful closure.
    b. The modified Cantwell-Ransley repair has replaced the Young technique because there is less urethral tortuosity and fistula rates are lower.
    c. The incidence of fistula formation was 12% at 3 months after epispadias repair.

d. Continence is more likely in those patients undergoing initial closure before 72 hours of age or those who have closure after 72 hours of age with osteotomy.
e. Continence rates are higher in those who have a capacity of 85 mL or more at the time of bladder neck reconstruction.

26. Which of the following statements is TRUE regarding exstrophy failures?
    a. After successful secondary closure, 90% of patients develop dryness and voided continence.
    b. Dehiscence after complete primary repair may be associated with corporal, urethral, and other major soft tissue loss.
    c. Bladder prolapse can be managed with minimal outlet procedures because this is considered a mild failure.
    d. Because the results of reclosure are poor, immediate resection of the bladder plate followed by neobladder construction is the preferred management.
    e. Posterior urethral stricture is usually a late complication occurring 4 to 6 years after initial closure.
27. Bladder neck reconstruction is designated as a failure if a 3-hour dry interval is not achieved within 2 years after surgery. Management of such failure is with the use of:
    a. collagen injection, which can lead to dryness.
    b. artificial urinary sphincter with small bladder capacity.
    c. bladder neck transection, augmentation cystoplasty, and continent diversion.
    d. repeat bladder neck reconstruction in relatively tight bladder necks.
    e. repeat bladder neck reconstruction in bladder instability.
28. The risks of ureterosigmoidostomy in the exstrophy population include:
    a. pyelonephritis and hyperchloremic acidosis.
    b. pyelonephritis, hyponatremia, and rectal incontinence.
    c. low incidence for eventual development of cancer.
    d. poor outcomes with upper tract deterioration.
    e. prolapse of the abdominal stoma.
29. What is the live birth incidence of cloacal exstrophy?

a. 1 in 20,000
b. 1 in 100,000
c. 1 in 400,000
d. 1 in 600,000
e. 1 in 1,000,000

30. All of the following statements regarding neurospinal abnormalities in cloacal exstrophy are true EXCEPT
    a. Thoracic defects may be noted in 10% of patients.
    b. The embryologic basis for the neurospinal defect has been identified as failure of neural tube closure.
    c. Autonomic bladder innervation is derived from a more medial location.
    d. Innervation of the duplicated corporal bodies arises from the sacral plexus and courses medial to the hemibladders.
    e. Functional defects can include minimal lower extremity function.
31. Which of the following is TRUE regarding cloacal exstrophy?
    a. The bones in a child with cloacal exstrophy are markedly different microscopically from those of healthy controls.
    b. In the presence of a normal bowel length, there is low probability for the development of short-gut syndrome.
    c. The most common Müllerian anomaly noted is partial uterine duplication.
    d. Cardiovascular and pulmonary anomalies are frequently noted.
    e. The most common upper urinary tract anomaly noted is multicystic dysplastic kidney.
32. What is the incidence of omphalocele associated with cloacal exstrophy?
    a. 20%
    b. 40%
    c. 60%
    d. 75%
    e. 95%
33. In the patient with cloacal exstrophy, hindgut remnants should be preserved in order to:

a. enlarge the bladder.
b. permit vaginal reconstruction.
c. allow either bladder augmentation or vaginal reconstruction.
d. provide additional length of bowel for fluid absorption.
e. allow later anal pull-through surgery.

34. Current research regarding gender in cloacal exstrophy indicates that:
    a. psychosexual evaluation indicates that patients have a marked female shift in development.
    b. patients have feminine childhood behavior but develop masculine gender identity.
    c. histology of the testis at birth is abnormal; therefore removal has been recommended.
    d. gender is based on quality of functional reconstruction rather than on karyotype.
    e. a functional and cosmetically acceptable phallus can now be constructed.
35. What is the live birth incidence of male epispadias?
    a. 1 in 72,000
    b. 1 in 117,000
    c. 1 in 150,000
    d. 1 in 250,000
    e. 1 in 400,000
36. What is the incidence of reflux in patients with complete epispadias?
    a. 10% to 20%
    b. 30% to 40%
    c. 50%
    d. 70%
    e. 90%
37. In the complete epispadias group, what is the predominant indicator of eventual continence?
    a. Length of the urethral groove
    b. Lack of spinal abnormalities
    c. Bladder capacity at the time of bladder neck reconstruction
    d. Age at bladder neck reconstruction

e. Age at epispadias repair and degree of resistance provided

38. All of the following are true regarding exstrophy-epispadias variants EXCEPT
    a. The presence of musculoskeletal defects characteristic of the complex, with a normal urinary tract, is termed *pseudoexstrophy.*
    b. The bladder is completely exstrophied in the superior vesical fissure variant.
    c. With "covered" exstrophy, an isolated ectopic bowel segment has frequently been noted.
    d. An isolated segment of bladder is left on the abdominal wall, with a complete urinary tract within the bladder in duplicate exstrophy.
    e. A common embryologic origin has been postulated for developments of all of the variants.
39. Sexual function and libido in male and female exstrophy patients are:
    a. normal in males, abnormal in females.
    b. normal only in males.
    c. normal in both males and females.
    d. normal only in females.
    e. abnormal in both males and females.
40. What is the most common complication after pregnancy in female exstrophy patients?
    a. Premature labor
    b. Rectal prolapse
    c. Preeclampsia
    d. Cervical and uterine prolapse
    e. Oligohydramnios
41. Psychologic studies of male and female children with bladder exstrophy find that:
    a. all have clinical psychopathology.
    b. they do not have clinical psychopathology.
    c. most have significant depression because of the condition.
    d. many children have gender dysphoria.

e. half of males and half of females have clinical psychopathology.

42. Single-stage reconstruction by using the complete primary exstrophy repair technique offers several advantages versus staged reconstruction EXCEPT
    a. The possibility of correcting the penile, bladder, and bladder neck abnormalities of bladder exstrophy with one operation.
    b. The ability to achieve urinary continence without bladder neck reconstruction.
    c. Correction of vesicoureteral reflux at the time of surgery.
    d. Lower complication rates than previous attempts at single-stage reconstruction.
    e. Initiation of bladder cycling early in life.
43. Single-stage reconstruction by using the complete primary exstrophy repair technique relies on which of the following to achieve continence?
    a. Reestablishment of normal anatomic relationships
    b. Bladder neck reconstruction at the time of primary surgery
    c. Osteotomy at the time of single-stage reconstruction
    d. Simultaneous epispadias repair
    e. None of the above
44. The following postoperative factors have been shown to increase the success of reconstruction for bladder exstrophy EXCEPT
    a. Immobilization with external fixators, Buck traction, a spica cast, or a mummy wrap
    b. Antibiotic therapy
    c. Prolonged status of nothing by mouth (NPO) to avoid abdominal distention
    d. Urinary diversion through ureteral stenting and suprapubic urinary drainage
    e. Adequate nutritional support
45. Single-stage reconstruction by using the complete primary exstrophy repair technique can be safely performed because:
    a. the neurovascular bundles of the corpus cavernosal bodies lie laterally rather than dorsally on the bodies.

b. the cavernosal bodies and urethral wedge are not actually separated from each other in this technique.
c. the blood supply to the cavernosal bodies and that to the urethral wedge are independent of each other.
d. the blood supply is quickly reestablished once the components are "reassembled."
e. the distal vascular communications between the corpora and urethral wedge are preserved.

46. The proximal limit(s) of dissection by using the complete primary exstrophy repair technique is/are:
    a. the intersymphyseal band.
    b. the muscles of the pelvic floor.
    c. the rectum.
    d. the corpora spongiosa.
    e. the endopelvic fascia.
47. Factors that mitigate against use of a single-stage reconstruction technique for cloacal exstrophy include the presence of:
    a. a large omphalocele.
    b. a wide pubic diastasis.
    c. a concomitant myelomeningocele.
    d. a small bladder plate.
    e. all of the above.
48. Complications of the complete primary exstrophy repair technique include:
    a. myogenic bladder failure.
    b. testicular atrophy.
    c. urethrocutaneous fistula.
    d. hip dislocation.
    e. epispadias.

Answers

1. **c. 1 in 50,000.** The incidence of bladder exstrophy has been estimated as between 1 in 10,000 and 1 in 50,000 live births.

2. **a. 1 in 70. Shapiro determined that the risk of bladder exstrophy in the offspring of individuals with bladder exstrophy and epispadias is 1 in 70 live births, a 500-fold greater incidence than in the general population.**
3. **b. Overdevelopment of the cloacal membrane, preventing medial migration of the mesodermal tissue and proper lower abdominal wall development. The theory of embryonic maldevelopment in exstrophy held by Marshall and Muecke is that the basic defect is an abnormal overdevelopment of the cloacal membrane. This theory of embryonic maldevelopment was held by Marshall and Muecke to explain the basic defect in exstrophy patients.**
4. **e. All of the above.** Sponseller and colleagues found that patients with classic bladder exstrophy have a mean external rotation of the posterior aspect of the pelvis of 12 degrees on each side, retroversion of the acetabulum, and a mean 18-degree external rotation of the anterior pelvis along with 30% shortening of the pubic rami.
5. **c. They are noted in 80% of boys and 10% of girls.** Connelly and colleagues, in a review of 181 children with bladder exstrophy, reported inguinal hernias in 81.8% of boys and 10.5% of girls.
6. **d. The anterior corporeal segments are 50% shorter than those of healthy control participants. With the use of magnetic resonance imaging (MRI) to examine adult men with bladder exstrophy and comparison of these results with those from age- and race-matched control participants, it was found that the anterior corporeal length in male patients with bladder exstrophy is almost 50% shorter than that in healthy control participants.**
7. **a. Volume weight and the cross-sectional area appeared healthy compared with published results from control participants.** The volume, weight, and maximal cross-sectional area of the prostate appeared normal compared with published results from control subjects.
8. **a. It is shorter than normal and of smaller caliber.** The vagina is shorter than normal, hardly greater than 6 cm in depth, but of

normal caliber.

9. **a. The density and binding affinity of the muscarinic receptors were similar to norms.** Muscarinic cholinergic receptor density and binding affinity were measured in control participants and in patients with classic bladder exstrophy. The density of the muscarinic cholinergic receptors in both the control and exstrophy groups was similar, as was the binding affinity of the muscarinic receptor. Therefore it was thought by the authors that the neurophysiologic composition of the exstrophied bladder is not grossly altered during its anomalous development.
10. **b. Before bladder neck reconstruction, 80% of patients had compliant and stable bladders.** Diamond and colleagues (1999), looking at 30 patients with bladder exstrophy at various stages of reconstruction, found that 80% of patients had compliant and stable bladders before bladder neck reconstruction.
11. **e. All of the above.** In a review of 25 prenatal ultrasonographic examinations with the resulting birth of a newborn with classic bladder exstrophy, several observations were made: (1) absence of bladder filling; (2) a low-set umbilicus; (3) a widened pubis ramus; (4) diminutive genitalia; and (5) a lower abdominal mass that increases in size while the pregnancy progresses and as the intra-abdominal viscera increase in size.
12. **c. Indentation of the bladder under anesthesia or outward bulging when the child cries.** In minor grades of exstrophy that approach the condition of complete epispadias with incontinence, the bladder may be small yet may demonstrate acceptable capacity, either by bulging when the baby cries or by indenting easily when touched by a sterile gloved finger in the operating room with the child under anesthesia.
13. **d. Bladder neck reconstruction before the epispadias repair to provide early continence.** The most significant changes in the management of bladder exstrophy have been (1) early bladder, posterior urethral, and abdominal wall closure, usually with osteotomy; (2) early epispadias repair; (3) reconstruction of a continent bladder neck and reimplantation of the ureters; and (4) definition of strict criteria for the selection of patients suitable for

this approach. **Bladder neck repair usually occurs when the child is 4 to 5 years of age, has an adequate bladder capacity, and, most important, is ready to participate in a postoperative voiding program.**

14. **c. Delaying closure by age 4 to 6 months with reassessment to see if the bladder will grow.** Ideally, waiting for the bladder template to grow for 4 to 6 months in the child with a small bladder is not as risky as submitting a small bladder template to closure in an inappropriate setting, resulting in dehiscence and allowing the fate of the bladder to be sealed at that point.
15. **c. There is no need to turn the patient to perform an osteotomy.** Combined osteotomy was developed for three reasons: (1) osteotomy is performed with the patient in the supine position, as is the urologic repair, thereby avoiding the need to turn the patient; (2) the anterior approach to this osteotomy allows placement of an external fixator device and intrafragmentary pins under direct vision; and (3) the cosmetic appearance of this osteotomy is superior to that of the posterior iliac approach.
16. **c. High rates of failure of reconstruction associated with the use of osteotomy and external fixation.** Successful closure was noted in 97% of those immobilized with an external fixator and modified Buck traction.
17. **b. The Erlangen approach includes all of the features of reconstruction of the exstrophy in a single procedure.** This method is truly a “complete repair” because it accomplishes all of the facets of exstrophy reconstruction in a single procedure. Surgical repair is, however, performed at 8 to 10 weeks of age, when the infant is larger and has had the opportunity to be medically stabilized.
18. **e. Cystoscopy. An important caveat is that if there are recurrent urinary tract infections or if the bladder is distended on an ultrasonographic study, cystoscopy should be performed and the posterior urethra should be carefully examined anteriorly for erosion of the intrapubic stitch, which may be the cause of the recurrent infections.**

19. **c. Intermittent antibiotics for urinary tract infections.** Before removal of the suprapubic tube 4 weeks after surgery, the bladder outlet is calibrated by a urethral catheter or a urethral sound to ensure free drainage. A complete ultrasound examination is obtained to ascertain the status of the renal pelves and ureters, and appropriate urinary antibiotics are administered because all patients will have reflux postclosure. Residual urine is estimated by clamping the suprapubic tube, and specimens for culture are obtained before the patient leaves the hospital and at subsequent intervals to detect infection and ensure that the bladder is empty.
20. **d. 60%.** In one study, if a patient underwent two closures, the chance of having an adequate bladder capacity for bladder neck reconstruction was 60%.
21. **a. Correction of ventral chordee.** Regardless of the surgical technique chosen for reconstruction of the penis in bladder exstrophy, four key concerns must be addressed to ensure a functional and cosmetically pleasing penis: (1) correction of dorsal chordee, (2) urethral reconstruction, (3) glanular reconstruction, and (4) penile skin closure.
22. **d. Bladder capacity. The most important long-term factor gleaned from a review of all these series is the fact that bladder capacity at the time of bladder neck reconstruction is an important determinant of eventual success.**
23. **c. 1 year.** The vast majority of patients achieve daytime continence in the first year after bladder neck reconstruction.
24. **b. 6 months.** Dehiscence—which may be precipitated by incomplete mobilization of the pelvic diaphragm and inadequate pelvic immobilization postoperatively, wound infection, abdominal distention, or urinary tube malfunction—necessitates a 6-month recovery period before a second attempt at closure.
25. **a. The onset of eventual continence and continence rates were unchanged in those who had initial successful closure. The importance of a successful initial closure is emphasized by Oesterling and Jeffs (1987) and by Husmann and colleagues (1989), who found that the onset of eventual continence was**

quicker and the continence rate higher in those who underwent a successful initial closure with or without osteotomy.

26. **b. Dehiscence after complete primary repair may be associated with corporeal, urethral, and other major soft tissue loss.** Dehiscence and prolapse have also been reported after the "complete repair" and may be associated with glandular, corporeal, urethral plate, and other major soft tissue loss.
27. **c. Bladder neck transection, augmentation cystoplasty, and continent diversion. A majority of bladder neck failures require eventual augmentation or continent diversion.** The artificial urinary sphincter has been used with some success in patients who have a good bladder capacity. However, in most of these failures the bladder capacity is small and augmentation will be required. At the time of reoperative surgery, the bladder neck is transected proximal to the prostate with a Mitrofanoff substitution, a continence procedure such as an artificial sphincter or collagen injection is performed, or both are performed. In our extensive experience with failed bladder neck reconstruction, most of the patients have had several surgeries and must be dry. In such cases the most suitable alternative is bladder neck transection, augmentation, and a continent urinary stoma (Gearhart et al., 1995b; Hensle et al., 1995).
28. **a. Pyelonephritis and hyperchloremic acidosis.** However, this form of diversion should not be offered until one is certain that anal continence is normal and after the family has been made aware of the potentially serious complications including pyelonephritis, hyperchloremic acidosis, rectal incontinence, ureteral obstruction, and delayed development of malignancy.
29. **c. 1 in 400,000.** Fortunately cloacal exstrophy is exceedingly rare, occurring in 1 in 200,000 to 400,000 live births.
30. **b. The embryologic basis for the neurospinal defect has been identified as failure of neural tube closure.** The embryologic basis for the neurospinal defects associated with cloacal exstrophy has been postulated to be secondary to problems with the disruption of the tissue of the dorsal mesenchyme rather than failure of neural tube closure (McLaughlin et al., 1995).

Alternatively, it has been suggested that the defects that lead to the formation of cloacal exstrophy may lead to the developing spinal cord and vertebrae being pulled apart (Cohen, 1991).

31. **c. The most common Müllerian anomaly noted was partial uterine duplication.** This anomaly is reported to be as high as 95% (Diamond, 1990). The vast majority of these patients had partial uterine duplication, predominantly a bicornate uterus.
32. **e. 95%.** In Diamond's series, the incidence of omphalocele was 88%, with a majority of all series reporting 95% or greater.
33. **d. Provide additional length of bowel for fluid absorption.** With the recognition of the metabolic changes in patients with ileostomy, an attempt is always made to use the hindgut remnant to provide additional length of bowel for fluid absorption.
34. **e. A functional and cosmetically acceptable phallus can now be constructed. Most authors recommend assigning gender consistent with the karyotypic makeup of the individual if at all possible.** This policy can be supported by a report indicating that the histology of the testis at birth is normal (Mathews et al., 1999a). Furthermore, with the evolution of techniques for phallic reconstruction, a functional and cosmetically acceptable phallus can now be constructed (Husmann et al., 1989).
35. **b. 1 in 117,000.** Male epispadias is a rare anomaly, with a reported incidence of 1 in 117,000 males.
36. **b. 30% to 40%.** The ureterovesical junction is inherently deficient in complete epispadias, and reflux has been reported between 30% and 40% in a number of series.
37. **c. Bladder capacity at the time of bladder neck reconstruction.** In the epispadias group, much as in the exstrophy group, bladder capacity is the predominant indicator of eventual continence.
38. **b. The bladder is completely exstrophied in the superior vesical fissure variant.** In the superior vesical fissure variant of the exstrophy complex, the musculature and skeletal defects are exactly the same as those in classic exstrophy; however, the persistent cloacal membrane ruptures only at the uppermost portion, and a superior vesical fistula that actually resembles a

vesicostomy results. Bladder extrusion is minimal and is present only over the normal umbilicus.

39. **c. Normal in both males and females.** Sexual function and libido in exstrophy patients are normal.
40. **d. Cervical and uterine prolapse.** The main complication after pregnancy was cervical and uterine prolapse, which occurred frequently.
41. **b. They do not have clinical psychopathology.** The conclusion of this long-term study was that children with exstrophy do not have clinical psychopathology.
42. **c. Correction of vesicoureteral reflux at the time of surgery.** In most applications of the primary exstrophy repair technique, correction of vesicoureteral reflux is not performed, although some have reported performing ureteral reimplantation. All of the other elements are considered advantages of the primary repair.
43. **a. Reestablishment of normal anatomic relationships.** The fundamental basis of the primary repair technique is to reposition the bladder neck and urethral complex into the normal pelvic position more posteriorly than at birth. This permits more normal function of the pelvic floor in maintenance of continence. The other factors do not contribute as significantly to continence.
44. **c. Prolonged status of nothing by mouth (NPO) to avoid abdominal distention.** It is not necessary to maintain an NPO status after primary repair because this will compromise nutrition. If an ileus develops, appropriate decompression and management are necessary because abdominal distention strains the repair. All other factors contribute to a successful outcome.
45. **c. The blood supply to the corpus cavernosal bodies and that to the urethral wedge are independent of each other.** Because the three elements of the penis, the two corpora, and the urethral wedge are fully separated in the penile disassembly, their vasculature must be proximal, which it is; that accounts for the success of this method. Nevertheless, preservation of these proximal vascular supplies is essential.
46. **b. The muscles of the pelvic floor.** The limit of dissection along the penile structures is the pelvic floor, which is then split to permit

repositioning of the bladder neck complex posteriorly.

47. **e. All of the above.** All of these factors would indicate that an attempt to perform a primary repair would be at high risk for failure, predominantly by dehiscence. Several of these factors may be present at one time.
48. **c. Urethrocutaneous fistula.** The most common complication after primary repair is development of a urethrocutaneous fistula on the ventrum of the penis. Other complications can include corporeal devascularization, hydronephrosis, and hypospadias.

Chapter Review

1. The male-to-female ratio for exstrophy is 2.3:1.
2. The risk of bladder exstrophy in family members is increased.
3. Rectal prolapse frequently occurs in untreated exstrophy patients who have widely separated symphyses. It disappears after exstrophy closure.
4. If rectal prolapse occurs after closure, bladder outlet obstruction should be suspected.
5. The autonomic nerves are displaced laterally in patients with exstrophy.
6. Reflux occurs in 100% of patients with exstrophy; inguinal hernias are common.
7. An ectopic isolated bowel segment may be present in the lower abdominal wall.
8. Osteotomy is rarely performed in newborns unless the diastasis is greater than 4 cm.
9. The most reliable predictors of urinary continence are the size of the bladder template at birth and successful primary closure.
10. Approximately 75% of patients with exstrophy are continent after repair. Continence is defined as 3 hours of dryness.
11. Cloacal exstrophy consists of exstrophy of the bladder; complete phallic separation; wide diastasis of the pubis; exstrophy of the terminal ileum, which lies between the two halves of the bladder; rudimentary hindgut; imperforate anus; omphalocele; and, not

infrequently, associated spinal defects. Spinal defects are not common in patients who have only exstrophy.

12. In adolescents and adults with exstrophy, concerns in the male are length, appearance, and deviation of the penis. In the female, concerns are the appearance of the external genitalia, adequacy of the vaginal opening, and uterine prolapse.
13. Women with exstrophy have delivered children; however, a frequent complication after pregnancy is cervical and uterine prolapse.
14. Closure of exstrophy: (1) reshapes the pelvis, (2) redistributes the levator group, and (3) smooths the contour of the pelvic floor.
15. At birth the exstrophy patient should have the umbilical cord secured with silk rather than an umbilical clamp to prevent trauma to the exposed bladder, and the bladder should be covered with a nonadherent film to minimize trauma and prevent desiccation of the bladder mucosa.
16. Bladder spasms must be controlled in the postoperative period following closure of the exstrophy.
17. Patients with epispadias may have associated vesicoureteral reflux and inguinal hernias, although the incidence is not as high as it is in bladder exstrophy.
18. The factor most likely to cause long-term disability in the reconstructed cloacal exstrophy patient is the associated neurologic deficit.
19. The basic defect in exstrophy is an abnormal overdevelopment of the cloacal membrane, preventing medial migration of the mesenchymal tissue and proper lower abdominal wall development.
20. Anterior corporal length in male patients with bladder exstrophy is almost 50% shorter than that of healthy control participants.
21. Bladder neck repair usually occurs when the child is 4 to 5 years old; has an adequate bladder capacity; and, most important, is ready to participate in a postoperative voiding program.
22. In a closed exstrophy patient, recurrent urinary tract infections should prompt evaluation for erosion of the anterior pubic stitch into the bladder or urethra.
23. A majority of bladder neck failures require eventual augmentation or continent diversion.

24. At birth, most recommend assigning gender that is consistent with karyotypic makeup of the individual if at all possible.

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[a] Sources referenced can be found in *Campbell-Walsh Urology*, *12th Edition*, on the Expert Consult website.

32

Prune-Belly Syndrome

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Keywords

cryptorchidism; prune-belly syndrome; ureterohydronephrosis; urinary tract reconstruction

Questions

1. The common antenatal ultrasonographic findings of a fetus with prune-belly syndrome are all of the following EXCEPT:
    a. oligohydramnios.
    b. empty scrotal sac.
    c. hydroureteronephrosis.
    d. distended bladder.
    e. ambiguous genitalia.
2. Which of the following is the most appropriate indication for antenatal intervention in a fetus with prune-belly syndrome?
    a. Distended bladder
    b. Bilateral hydroureteronephrosis
    c. Pulmonary hypoplasia
    d. Progressive oligohydramnios
    e. Urinary ascites
3. Common findings in the urinary tract of patients with prune-belly syndrome include all of the following EXCEPT:
    a. renal dysplasia.

b. ureteropyelocaliectasis.
c. vesicoureteral reflux.
d. elevated bladder neck.
e. urachal diverticulum.

4. Common nonurinary findings in patients with prune-belly syndrome include all of the following EXCEPT:
   a. flaccid abdomen due to nonhomogeneous deficiency of the abdominal musculature.
   b. intestinal obstruction.
   c. impalpable testes.
   d. pectus carinatum.
   e. pulmonary deficiency.
5. Which of the following organ systems are MOST likely to impact the early life of the patient with prune-belly syndrome?
   a. Cardiac
   b. Urinary
   c. Pulmonary
   d. Endocrine
   e. Orthopedic
6. Considering the prognosis of untreated patients with prune-belly syndrome, which statement is FALSE?
   a. Most patients present some degree of urinary tract dilatation.
   b. Recurrent urinary tract infection increases the risk of renal damage.
   c. The intra-abdominal testes may descend spontaneously before puberty.
   d. Abdominal wall laxity may be associated with ineffective cough and respiratory illnesses.
   e. Bladder emptying may improve spontaneously with time.
7. Considering the dilatation of the upper urinary tract of patients with prune-belly syndrome, which of the following statements is TRUE?
   a. The degree of dilation is not proportional to the abdominal wall laxity.
   b. Pyelocaliceal dilatation is never associated with ureteropelvic junction obstruction.

c. Histologically, the thickened ureteral wall has an increased number of muscular cells.
d. Ureteral dilation is proportional to the degree of vesicoureteral reflux.
e. All statements above are correct.

8. Considering the lower urinary tract of patients with prune-belly syndrome, which of the following statements is TRUE?
   a. The thickened and enlarged bladder has poor compliance and uninhibited bladder contractions.
   b. Elevated bladder pressures occur due to detrusor-sphincter dyssynergia.
   c. Large postvoid residuals occur due to poorly contractile detrusor and diminished bladder sensation.
   d. Reduction cystoplasty reduces permanently the postvoid residual.
   e. All statements above are correct.
9. Considering the characteristics of the urethra of prune-belly syndrome patients, which of the following statements is TRUE?
   a. The posterior urethra is dilated due to distal obstruction.
   b. The lack of bladder neck hypertrophy is typical of the posterior urethral dilatation.
   c. Megalourethra is a frequent finding and is due to distal obstruction.
   d. Hypospadias is commonly associated.
   e. Prostatic hyperplasia is associated with upper tract dilatation.
10. Regarding abdominal undescended testes in prune-belly syndrome, which of the following statements is CORRECT?
   a. The testes must be brought down to the scrotum at the age of 2 years if spontaneous descent hasn't occurred before then.
   b. Due to the position of the testes, the Fowler-Stephens technique is always necessary.
   c. In cases of successful orchidopexy, the testes maintain endocrine function but are incapable of preserving spermatogenesis.
   d. Most patients require hormonal supplementation in adulthood.
   e. None of the statements are correct.

11. Considering the management of patients with prune-belly syndrome, which of the statements is CORRECT?
    a. The spontaneous improvement of the urinary tract dilatation with time is the basis for the clinical management, particularly in grade III prune belly syndrome (PBS) patients.
    b. In newborns with significant dilatation and vesicoureteral reflux, early ureterostomy is always indicated.
    c. Continuous antibiotic prophylaxis and antimuscarinic drugs are necessary in all patients, to preserve the upper urinary tract.
    d. Only patients with persistently elevated postvoid residuals must undergo internal urethrotomy.
    e. All statements are correct.
12. Considering the surgical management of PBS patients, which of the statements is CORRECT?
    a. Abdominoplasty must be planned and performed after urinary tract reconstruction is completed.
    b. The surgical and anesthesiologic risks are elevated when orchidopexy and abdominoplasty are performed simultaneously with the urinary tract reconstruction before puberty.
    c. Vesicoureteral reflux can be corrected with the Gregoir-Lich procedure in most cases.
    d. Even when reduction cystoplasty is performed, significant postvoid results may remain or recur.
    e. All statements are false.

Answers

1. **e. Ambiguous genitalia.** Although not present in all cases, the common findings in the antenatal ultrasound in prune-belly syndrome are urinary tract dilatation with a large, full bladder, and abnormal abdominal circumference. The penis is normal to fetal age, but the scrotum is empty due to abdominal cryptorchidism.

There are no reports of association of PBS with ambiguous genitalia.

2. **d. Progressive oligohydramnios.** Bilateral hydronephrosis with a distended bladder is usually well tolerated and does not represent a definitely obstructed urinary tract. Therefore, no antenatal intervention is necessary. The presence of isolated urinary ascites or pulmonary hypoplasia also does not warrant drainage of the fetal bladder. Nevertheless, the association of those findings with a progressively decreased amniotic fluid most probably represents an obstructed bladder due to an atretic urethra associated with functioning kidneys, which can be treated by bladder decompression.
3. **d. Elevated bladder neck.** In prune-belly syndrome, the kidneys may present variable degrees of renal dysplasia, with substitution of the normal parenchyma by cystic structures. This may affect one or both kidneys, in a pattern that is not proportional to the patient's abdominal flaccidity, the degree of dilatation of the urinary tract, or the presence of vesicoureteral reflux. When bilateral dysplasia occurs, survival is compromised. Hydroureteronephrosis is also variable, but in general does not impact in the initial renal function. Vesicoureteral reflux occurs in two-thirds of the patients, eventually associated with more dilated lower ureteral segments. Urachal diverticulum is present frequently, usually with thick musculo-fibrotic walls. In more rare cases, associated with an atretic urethra, the urachus can be completely patent to the umbilicus, in this case functioning as an efficient bladder drainage that helps preserve the upper urinary tract function. The posterior urethra is dilated, but contrary to patients with posterior urethral valves, the bladder neck is wide open, making it difficult to characterize the transition between the posterior urethra and the base of the bladder.
4. **b. Intestinal obstruction.** The abdominal flaccidity is the hallmark of the syndrome. It affects unevenly the medial and inferior segments of the abdomen, but in some patients the lateral aspects are more compromised. The musculo-aponeurotic layer of the abdominal wall is very deficient, with a very thin layer of muscle cells interwoven with aponeurotic tissue that hardly exert any

contraction and support to respiration, therefore impairing pulmonary function. The lack of adequate abdominal muscular anchoring and support is responsible for the cases that present with pectus carinatum or scavatum. The lack of intra-abdominal pressure due to the weakness of the abdominal wall musculature is one of the causes of abdominal cryptorchidism. Despite eventually described intestinal abnormalities, obstruction doesn't occur primarily, being reported as a complication of intraperitoneal surgical procedures.

5. **c. Pulmonary.** Significant cardiac, orthopedic, and endocrine abnormalities that impact early patient survival are infrequent. Urinary tract abnormalities may be significant, particularly if kidney function is impaired due to bilateral dysplasia. Hydroureteronephrosis, vesicoureteral reflux, and difficult bladder emptying are also important factors, but are rarely fatal in early infancy. More frequently, it is the pulmonary hypoplasia due to oligohydramnios, often aggravated by the lack of abdominal muscle support, that impacts survival, thereby requiring prolonged respiratory support.
6. **c. The intra-abdominal testes may descend spontaneously before puberty.** The abdominal testes are situated high in the intra-abdominal cavity, usually laying upon the dilated distal ureteral folds. The inguinal region and the scrotum are empty. There is no description of spontaneous descent of the testes to the inguinal or scrotal position in a patient with the full spectrum of prune-belly syndrome, and the lack of abdominal pressure is one of the factors that preclude this descent.
7. **a. The degree of dilation is not proportional to the abdominal wall laxity.** The dilation of the upper urinary tract can be significant in one or both sides but has no correlation with the intensity of abdominal wall flaccidity. There are cases where the transition from the renal pelvis to the upper ureter may be considerably narrow, due to thickened renal pelvic or ureteral walls secondary to increased collagen, suggesting stenosis. Sometimes the ureteral folds at this segment may also cause significant difficulty to the drainage of the renal pelvis. The lower ureteral

dilatation is usually more intense than that of the upper ureter, independent of the presence of vesicoureteral reflux.

8. **c. Large postvoid residuals occur due to poorly contractile detrusor and diminished bladder sensation.** The bladder usually has thick walls due to increase in collagen and a decrease in smooth muscle cells. Functionally, it has a large capacity with increased compliance but decreased sensitivity and poor contraction. Voiding pressures are usually low, and adequate bladder emptying is achieved only with the help of increased abdominal pressure. Significant postvoid residuals can be seen in many patients. In the case of a very large bladder, a reduction cystoplasty can decrease the postvoid residual, but the long-term effects of this procedure are not consistent.
9. **b. The lack of bladder neck hypertrophy is typical of the posterior urethral dilatation.** The characteristic dilatation of the posterior urethra is mostly due to the lack of support by prostatic tissue, dilatation being present even in the presence of a normal distal urethra. The lack of bladder neck hypertrophy distinguishes prune-belly syndrome from posterior urethral valves, where increased bladder pressures are observed. The presence of megalourethra in the penile urethral segment is not associated with distal stenosis, but is due to the segmental lack of support by the spongiosum.
10. **e. None of the statements is correct.** As in nonsyndromic cryptorchidism, the adequate time for orchidopexy is between 6 and 18 months of age. There is no formal indication to anticipate or postpone the procedure in patients with prune-belly syndrome, except when another major procedure, such as abdominoplasty or urinary tract reconstruction, is being planned for an earlier or later period. As in all children with abdominal undescended testes, the orchidopexy can be successfully carried out without the Fowler-Stephens technique before the second year of age. In adult patients who underwent early successful orchidopexy, the hormonal function is normally preserved, with no need of hormonal supplementation. The presence of sperm in the ejaculate or

postmasturbation urine in adult patients has been well documented, and shows potential fertility with assisted reproduction techniques.

11. **a. The spontaneous improvement of the urinary tract dilatation with time is the basis for the clinical management, particularly in grade III prune belly syndrome (PBS) patients.** As in cases of primary megaureter and vesicoureteral reflux, a considerable functional and anatomic improvement of the upper urinary tract is expected. Continuous antibiotic prophylaxis must be employed in the more vulnerable patients to prevent pyelonephritic injuries to the kidneys. As there is no increased detrusor activity or hypertrophic bladder neck in these patients, there is no indication for the use of antimuscarinic drugs or alpha-blockers. Upper urinary tract diversion in cases of evident upper urinary tract obstruction is rarely indicated, as urinary tract reconstruction is indicated in these cases. Persistent and elevated postvoid residual must be treated by timed voiding and/or clean intermittent catheterization.
12. **d. Even when reduction cystoplasty is performed, significant postvoid results may remain or recur.** Whenever necessary, urinary tract reconstruction can be performed simultaneously with abdominoplasty, orchidopexy, and circumcision. It has been documented that the procedure has acceptable rates of surgical and anesthetic complications even in infants. The urinary tract reconstruction aims to decrease the ureteral redundancy by removing the distal, more dilated ureteral segments and reimplanting them in the bladder. At the same time, removal of the urachal diverticulum and part of the noncontractile bladder dome is usually performed, but it doesn't prevent the persistence or recurrence of significant postvoid residual.

Chapter Review

1. Hydroureteronephrosis is often to a severe degree; however, the calyceal morphology may be well preserved.

2. The proximal portion of the ureters has more normal muscle than the distal portions.
3. The bladder is large with a pseudodiverticulum at the urachus and a wide bladder neck opening into a dilated prostatic urethra.
4. Megalourethra is more commonly seen in PBS than any other syndrome. With the scaphoid variety, the ventral urethra dilates with voiding, whereas with the fusiform variety, the entire phallus dilates with voiding.
5. Infertility is thought to be due to a combination of testicular histologic abnormalities, structural defects of the ducts and prostatic abnormalities, and retrograde ejaculation.
6. Pulmonary hypoplasia can result from severe oligohydramnios related to renal dysplasia, or severe bladder outlet obstruction, and may result in newborn demise.
7. Despite being one of the hallmarks of the syndrome, the abdominal flaccidity has no direct relationship with the intensity of the urologic abnormalities and doesn't impair the short- and long-term survival of the patient. Together with the orthopedic abnormalities, if significant and left untreated, it may cause a decrease in the quality of life in the adolescent and adult patients.
8. A team consisting of neonatology, nephrology, urology, and other specialties such as cardiology is necessary.
9. A voiding cystourethrogram (VCUG) is indicated in the neonatal period if there is renal insufficiency or evidence of bladder outlet obstruction and only after antibiotic prophylaxis.
10. A chest x-ray to evaluate for pneumothorax, pneumomediastinum, and pulmonary hypoplasia is necessary.
11. Baseline assessment of renal function should include renal and bladder ultrasonography, blood urea nitrogen (BUN), creatinine, and electrolytes.
12. Circumcision is advisable in the absence of a structural penile abnormality.
13. Early intervention is indicated for evidence of bladder outlet obstruction and preferably with a percutaneous suprapubic tube.
14. Upper urinary tract reconstruction is controversial but clearly indicated for evidence of declining renal function in the presence of

recurrent upper tract infections or progression of the hydroureteronephrosis.

15. Orchidopexy is best performed early in life because this affords the most likely prospect of a successful single-stage procedure, as well as the perspective of preservation of germ cells.
16. Abdominal wall reconstruction has demonstrated improved bladder emptying, a more effective cough, and improved defecation, though long-term utility for appearance is still debatable.

33

Posterior Urethral Valves

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Keywords

oligohydramnios; posterior urethral valves; urethral anomalies; valve ablation; valve bladder syndrome

Questions

1. A newborn infant with a history of antenatal hydronephrosis and oligohydramnios is evaluated with a renal/bladder ultrasound and voiding cystourethrogram. All of the following are characteristic findings of posterior urethral valves EXCEPT:
    a. hypertrophy and apparent elevation of the bladder neck.
    b. multiple bladder diverticula
    c. bladder perforation with small amount of urinary extravasation causing ascites.
    d. rupture of an upper pole calyceal fornix causing distortion of the renal capsule.
    e. all of the above.
2. All of the following are true regarding antenatal diagnosis and management of posterior urethral valves EXCEPT:
    a. Thickened dilated bladder with bilateral upper tract dilation are pathognomonic findings.
    b. Significant renal cortical dysplasia is a contraindication to antenatal intervention.

c. Vesicoamniotic shunting has been shown to improve renal failure rates.
d. Early reports suggest that fetal cystoscopy and valve ablation might offer better outcomes than vesicoamniotic shunting.
e. Antenatal interventions should only be offered in select cases as outcomes are not universally predictable.

3. A 6-month-old child with a history of posterior urethral valves has a serum creatinine of 1.4 mg/dL. He has a dilated, thick-walled bladder and hydroureteronephrosis on sonography and evidence of dilating vesicoureteral reflux with no remnant valves on voiding cystourethrography. Which of the following BEST explains the renal impairment noted in this child?
    a. Increased intravesical storage pressures transmitted to the ureter, renal pelvis, and glomerular units causing architectural and functional changes
    b. High-grade vesicoureteral reflux
    c. Repeated afebrile urinary tract infections
    d. Family history
    e. Likely misdiagnosis of hydronephrosis for concurrent autosomal dominant polycystic kidney disease
4. A newborn diagnosed with posterior urethral valves is noted on VCUG to have unilateral high-grade vesicoureteral reflux into the right kidney that has no measurable renal function on nuclear renography. The left kidney displays normal uptake and excretion of nuclear tracer. Which of the following best characterizes this boy's renal status?
    a. Better long-term renal function due to protective effect of reflux into dysplastic right kidney
    b. No better or worse long-term renal function, still requiring close observation
    c. High risk of febrile urinary tract infections will require early nephroureterectomy of nonfunctioning renal unit
    d. Plan early right ureteral reimplantation to reduce risk of pyelonephritis
    e. None of the above

5. All of the following findings on antenatal imaging should raise suspicion of posterior urethral valves EXCEPT:
    a. thickened bladder wall.
    b. bilateral pelvicaliectasis with ureterectasis.
    c. oligohydramnios.
    d. ambiguous genitalia.
    e. dilated posterior urethra (keyhole sign).
6. The most common cause of early neonatal mortality in a baby affected by posterior urethral valves is:
    a. urinary sepsis.
    b. end-stage renal disease not amenable to dialysis.
    c. pulmonary hypoplasia.
    d. urinary ascites due to calyx forniceal rupture.
    e. necrotizing enterocolitis.
7. A 3-year-old circumcised boy presents with urinary incontinence, stranguria, and occasional afebrile urinary tract infections. A renal ultrasound reveals moderate bilateral hydroureteronephrosis. The next most appropriate urological evaluation for this child would be:
    a. ordering a repeat renal ultrasound in 6 months.
    b. treatment of dysfunctional elimination syndrome with timed voiding and laxative.
    c. beginning an alpha receptor blocker.
    d. Botox injection to the external urethral sphincter.
    e. voiding cystourethrogram.
8. Fetal intervention for obstructive uropathy secondary to posterior urethral valves has been shown to:
    a. diminish the incidence of end-stage renal disease.
    b. be associated with a high rate of fetal demise.
    c. lead to improved pulmonary function in the neonate.
    d. be most effective when accomplished by open fetal surgery.
    e. be associated with neonatal respiratory failure.
9. A premature neonate with a weight of 2500 g with impaired renal function and bilateral hydroureteronephrosis (right greater than left) is diagnosed with posterior urethral valves. After 1 week of bladder catheterization, a plan is made to proceed to the operating room for

valve ablation. Cystoscopy is precluded by the small genitalia. Which of the following is the preferred initial surgical option?

a. Creation of a vesicostomy
b. Bilateral proximal loop ureterostomy
c. Cystotomy with antegrade valve ablation
d. Suprapubic tube placement until child is old enough for valve ablation
e. Bilateral percutaneous nephroureteral stent placement

10. A 4-year-old boy with a history of ablation of posterior urethral valves in infancy is volitionally voiding with no incontinence, stable renal function, and no urinary tract infections. He has high-grade vesicoureteric reflux on the left side with stable hydroureteronephrosis that has not changed since infancy. What is the best management for this child?

a. Cross-trigonal ureteral reimplantation
b. Vesicostomy
c. Conservative management with timed voiding and anticholinergic therapy
d. Cystoscopy with subureteral injection of Deflux
e. Both a and d are correct

11. In cases of posterior urethral valves, the bladder transitions through three contractility patterns in childhood. Choose the correct order for the changes in bladder contractility.

a. Detrusor hyperreflexia with high intravesical pressure; improved compliance with reduced intravesical pressures; high-capacity bladder and hypocontractility
b. High-capacity bladder and hypocontractility; detrusor hyperreflexia with high intravesical pressure; improved compliance with reduced intravesical pressures
c. Improved compliance with reduced intravesical pressures; high-capacity bladder and hypocontractility; detrusor hyperreflexia with high intravesical pressure
d. High-capacity bladder and hypocontractility; improved compliance with reduced intravesical pressures; detrusor hyperreflexia with high intravesical pressure

e. Detrusor hyperreflexia with high intravesical pressure; high-capacity bladder and hypocontractility; improved compliance with reduced intravesical pressures

12. All of the following are associated with the development of the valve bladder syndrome EXCEPT:
    a. High voiding pressures
    b. Incomplete bladder emptying with high post-void residuals
    c. Renal tubular and glomerular impairment
    d. Previous history of vesicostomy
    e. Bilateral hydroureteronephrosis
13. A child with a history of posterior urethral valves ablation suffers renal impairment and, ultimately, progresses to end-stage renal disease. All of the factors below likely characterized his clinical course EXCEPT:
    a. nadir creatinine at 1 year of life.
    b. renal dysplasia with or without vesicoureteral reflux.
    c. recurrent urinary tract infections.
    d. valve bladder syndrome.
    e. all of the above.
14. A 13-year-old boy with a history of posterior urethral valves progresses to end-stage renal disease and is a candidate for renal transplant. Of the complications listed below, which one may be most likely to occur in a child with a history of posterior urethral valves?
    a. Acute graft rejection
    b. Vesicoureteral reflux
    c. Chronic graft rejection
    d. Ureteral obstruction at site of ureteroneocystostomy
    e. All of the above
15. In a child with urethral duplication, all of the following are true EXCEPT:
    a. In a case of duplicated urethra, the ventral urethra is morphologically more normal in terms of caliber and function.
    b. Not all urethral duplications require surgical correction.
    c. Most children with urethral duplication present with incontinence.

d. Children with urethral duplication are not at high risk for urinary tract infection.
e. Can occur in both genders.

16. An 8-year-old boy presents with a few drops of blood at the urethral meatus at the end of voiding intermittently, over the past 3 months. Which of the following is true regarding the diagnosis of urethrorrhagia?
    a. Presents classically as painful hematuria
    b. Associated with an increased risk of urinary tract infection
    c. Cystoscopy is essential to locate a bladder or urethral source of bleeding and rule out the presence of a neoplasm.
    d. May be associated with dysfunctional elimination syndrome
    e. Renal and bladder ultrasonography is diagnostic

Answers

1. **c. Bladder perforation with small amount of urinary extravasation causing ascites.** Fetal and neonatal lower urinary tract obstruction due to posterior urethral valves may transmit enough pressure to the upper urinary tract to cause a rupture of a calyceal fornix causing urine to be trapped within the renal capsule or cause ascites, in rare cases. However, bladder perforation is not a characteristic of posterior urethral valves, since the process actually leads to significant bladder wall hypertrophy.
2. **c. Vesicoamniotic shunting has been shown to improve renal failure rates.** Thickened dilated bladder and bilateral upper tract dilation imply posterior urethral valves with high sensitivity. Criteria for intervention include favorable urinary chemistry, a singleton pregnancy, normal karyotype, and absence of significant dysplasia. Antenatal intervention includes vesicoamniotic shunting and fetal cystoscopic ablation. Vesicoamniotic shunting is associated with an increased incidence of live births in the severely affected fetus, but is not correlated with decreasing the risk of renal failure. Early reports suggest fetal cystoscopy and ablation are more accurate in diagnosing valve patients and thus have better

outcomes. Experience in antenatal intervention is early and varied and hence should only be offered in highly selected cases.

3. **a. Increased intravesical storage pressures transmitted to the ureter, renal pelvis, and glomerular units causing architectural and functional changes.** Sustained increases in intravesical storage pressure over prolonged time intervals transmit these pressures to the ureter, renal pelvis, and, ultimately, glomerular units. Of the options provided, this is the most likely etiology of renal insufficiency in this child.
4. **b. No better or worse long-term renal function, still requiring close observation.** Contrary to previous assumptions, the so-called vesicoureteric reflux and dysplasia (VURD) syndrome is not a renal protective phenomenon. These children often have evidence of renal dysplasia detectable in the solitary functioning kidney, increasing the likelihood of significant long-term renal impairment.
5. **d. Ambiguous genitalia.** All of the aforementioned findings are characteristic findings of posterior urethral valves on antenatal imaging, except ambiguous genitalia. Posterior urethral valves are not associated with external genital anomalies.
6. **c. Pulmonary hypoplasia.** While the focus in posterior urethral valves is too often on the lower urinary tract and kidneys, the most profound complication and cause of perinatal mortality in infants affected by severe lower urinary tract obstruction remains pulmonary hypoplasia.
7. **e. Voiding cystourethrogram.** Despite the ubiquity of antenatal imaging today, delayed presentation of posterior urethral valves after 6 months of age is not uncommon. A high index of suspicion for posterior urethral valves must be assumed when a boy presents with lower urinary tract symptoms, especially recurrent urinary tract infections, but also overflow incontinence, gross hematuria, renal dysfunction, and, less commonly, ejaculatory dysfunction.
8. **c. Lead to improved pulmonary function in the neonate.** Fetal intervention for posterior urethral valves has not been shown to lower the incidence of end-stage renal disease, but, in properly selected cases, it can offer the benefit of improved neonatal

pulmonary function. Neonates with severe obstructive uropathy die of respiratory failure due to a noncompliant and hypoplastic lung.

9. **a. Creation of a vesicostomy.** In the low-birth-weight infant with renal impairment and whose urethra will not accommodate an infant cystoscope, a vesicostomy is the ideal first option. The vesicostomy allows decompression of the obstructed system, allows continued bladder cycling, and is easily managed with diapers. Upper tract diversion is a reasonable option, but does require bilateral incisions and also a complex follow-up surgery that can risk injury to the developing ureters.
10. **c. Conservative management with timed voiding and anticholinergic therapy.** Vesicoureteral reflux in children with posterior urethral valves is a common finding and should be understood to be a consequence of neonatal obstruction and the secondarily elevated bladder pressures. Management of this child should be centered on the lower urinary tract with attention to timed voiding, double voiding, and anticholinergics as necessary. Ureteral reimplantation is an option in atypical cases where urinary tract infections continue despite maximal bladder therapy.
11. **a. Detrusor hyperreflexia with high intravesical pressure; improved compliance with reduced intravesical pressures; high-capacity bladder and hypocontractility.** The bladder evolves through three distinct contractility patterns through childhood: (1) detrusor hyperreflexia in infancy and early childhood; (2) decreasing intravesical pressures and improved compliance bladder in childhood; (3) increased capacity bladder with hypocontractility and atony in adolescence.
12. **d. Previous history of vesicostomy.** The theory of the valve bladder syndrome holds that while the bladder initially compensates for outlet obstruction by generating high voiding pressures, it begins to experience higher volumes of urine due to increasing urine production as the child grows. The polyuria due to nephrogenic diabetes insipidus secondary to evolving renal impairment augments the urine volumes entering a bladder that is increasingly unable to empty completely. As the post-void residuals increase, the bladder no longer enjoys periods of complete

relaxation, and the detrusor fibers are continuously in a state of partial or complete stretch, beginning a cascade of gene expression and phenotypic changes that further impair contractility of the bladder. Creation of a vesicostomy does not impair bladder contractility.

13. **e. All of the above.** Risk factors known to affect the prognosis of an infant diagnosed with posterior urethral valves include age at diagnosis, renal dysplasia with or without vesicoureteral reflux, nadir creatinine during first year of life, recurrent urinary tract infections, and bladder dysfunction.
14. **d. Ureteral obstruction at site of ureteroneocystostomy.** The thickened bladder wall of posterior urethral valve patients may contribute to the significantly increased incidence of ureteral obstruction on univariate and multivariate analysis compared to a non–posterior urethral valve transplant cohort, but recent studies saw no risk of increased graft loss or patient death despite ureteral obstruction, stenting, or dilation.
15. **c. Most children with urethral duplication present with incontinence.** Most children with urethral duplication are continent since most duplications occur distal to the urethral sphincter. Although more common in boys, rare reports of female urethral duplications are seen in the literature. Urinary tract infection is not commonly seen in urethral duplication. Blind ending duplications or duplications where both urethra end in or near the orthotopic location do not need surgical correction. Clinical presentation with significant functional abnormalities require surgical reconstruction.
16. **d. May be associated with dysfunctional elimination syndrome.** Urethrorrhagia presents classically as bleeding at the end of urination, or with a blood stain in the underwear with no association of pain. It is not associated with urinary tract infection. Diagnosis is clinically rendered, based on history and examination and exclusion of other potential causes of gross hematuria. Cystoscopy is not essential for diagnosis, and there are no diagnostic radiological imaging exams. There is a significant association with voiding dysfunction, and improving voiding

parameters have been shown to improve symptoms and resolve urethrorrhagia.

Chapter Review

1. Type 1 valves are the most common variant of posterior urethral valves, and appear as leaflets that arise from the verumontanum and fuse anteriorly just proximal to the external urethral sphincter.
2. Type 3 valves present as a congenitally obstructing membrane that is likely perforated at the time of the initial post-natal catheterization.
3. The incidence of posterior urethral valves is between 1.6 and 2.1 per 10,000 births.
4. There is an emerging understanding of the genetic basis for posterior urethral valves, particularly associated with chromosome 11 and copy number variants.
5. The renal dysfunction, vesicoureteral reflux, and voiding dysfunction seen in children with posterior urethral valves is mediated by a dysfunctional bladder.
6. Renal impairment in posterior urethral valve patients is due to renal dysplasia and obstructive uropathy.
7. The vesicoureteral reflux and dysplasia (VURD) syndrome confers no protective benefit on long-term renal prognosis. Antenatal diagnosis is primarily by fetal ultrasound sometimes aided by fetal MRI. Combination of thickened dilated bladder with upper tract dilation along with oligohydramnios has a high sensitivity for posterior urethral valves.
8. Postnatal diagnosis is with ultrasound and finding confirmed by fluoroscopic voiding cystourethrogram. More recently, contrast-enhanced voiding ultrasonogram is emerging as a good alternative.
9. Biochemical evaluation centers on electrolyte and creatinine values. Nadir creatinine value is an effective prognostic indicator for long-term kidney function.
10. Up to 50% of infants with posterior urethral valves are diagnosed in the antenatal period.

11. Severity of obstruction can be objectively assessed by the volume of amniotic fluid, degree of renal dysplasia, and fetal urinary markers.
12. Vesicoamniotic shunting may have a survival benefit for the infant in select cases, but there is no clear benefit in the risk for renal failure.
13. Antenatal finding of a thickened bladder and bilateral ureterectasis should be evaluated with early postnatal ultrasound and voiding cystourethrogram.
14. VCUG in valve patient will show bladder wall irregularity, hypertrophied, elevated bladder neck, and dilated and elongated posterior urethra.
15. The most common cause of early mortality in infants with posterior urethral valves is pulmonary hypoplasia.
16. Cystoscopy with valve ablation is the preferred initial treatment for posterior urethral valves.
17. Vesicostomy does not inhibit bladder cycling, as the bladder continues to contract, but is reserved for select cases where valve ablation is not possible.
18. High urinary diversion offers no renal protective benefit and requires a complex secondary procedure to reverse the diversion.
19. Circumcision should be encouraged as a prophylactic measure for any boy with posterior urethral valves, and especially any boy with a history of urinary tract infection.
20. The focus of management for vesicoureteral reflux in a child with valves should be centered on the bladder, and ureteral reimplantation is rarely offered.
21. The bladder evolves through three patterns in boys with valves: (1) detrusor hyperreflexia in infancy and early childhood; (2) decreasing intravesical pressures and improved compliance bladder in childhood; (3) increased capacity bladder with hypocontractility and atony in adolescence.
22. Videourodynamics is completed in boys with valves especially when there is a clinical change, such as increasing upper tract dilation or recurrent urinary tract infections.
23. Overnight bladder drainage is considered in the scenario of increasing post-void residuals, urinary tract infections, or worsening hydronephrosis and renal function.

24. An antenatal vesicoamniotic shunt can be considered in select patients with bladder wall thickening, hydroureteronephrosis, and oligohydramnios. While it may reduce the severity of pulmonary hypoplasia, the procedure confers no protection from renal impairment.
25. A nadir serum creatinine at 1 year of less than 0.8 mg/dL confers a significantly decreased risk of developing end-stage renal disease.
26. Transplantation into a valve-affected bladder may carry a higher risk of ureteral obstruction, but there is no increased risk of graft loss compared to controls.
27. Anterior urethral valves and urethral atresia are less common causes of lower urinary tract obstruction (LUTO), and long-term outcomes can be similar to those of posterior urethral valves if the neonate survives.
28. Dysfunctional voiding causing a turbulent urinary flow may lead to urethrorrhagia.
29. Treatment for urethrorrhagia should focus on improving voiding habits and conservative management, reserving cystoscopy for refractory cases, and when there is significant straining during voiding.
30. An infant diagnosed with anorectal malformation should undergo a renal ultrasound as well as VCUG due to the high risk of associated genitourinary anomalies.

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Neuromuscular Dysfunction of the Lower Urinary Tract in Children

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Keywords

neurogenic bladder; neuromuscular dysfunction

Questions

1. Which of the following is an acquired form of neuromuscular dysfunction of the lower urinary tract?
    a. Myelomeningocele
    b. Cerebral palsy
    c. Lipomeningocele
    d. Sacral agenesis
    e. Anorectal malformation
2. What is the primary goal in management of neuromuscular dysfunction of the lower urinary tract?
    a. Achievement of urinary continence
    b. Achievement of fecal continence
    c. Preservation of renal function
    d. Facilitation of sexual function
    e. Avoidance of urinary tract infection
3. The International Children's Continence Society (ICCS) recommends more frequent evaluation of children during periods of

high rates of somatic growth when spinal cord tethering is more likely. These two development periods are:

a. newborn to toddler AND toddler to adolescent.
b. newborn to toddler AND adolescent to adult.
c. newborn to toddler AND adulthood.
d. toddler to adolescent AND adolescent to adult.
e. toddler to adolescent AND adulthood.

4. The ICCS's indications for repeat investigations prior the routinely scheduled follow-up for neuromuscular dysfunction of the lower urinary tract do NOT include:
    a. urinary tract infections.
    b. development or worsening of hydronephrosis.
    c. worsening continence.
    d. change in lower extremity function.
    e. improved continence.
5. Which of the following may compromise bladder emptying in neuromuscular dysfunction of the lower urinary tract?
    a. Low capacity
    b. Low compliance
    c. Detrusor overactivity
    d. Detrusor sphincter dyssynergia
    e. Low outlet resistance
6. Initial minimally invasive treatment options to address inadequate bladder storage in neuromuscular dysfunction in children usually involves:
    a. overnight indwelling catheter drainage.
    b. antimuscarinics and clean intermittent catheterization (CIC).
    c. percutaneous cystostomy tube.
    d. endoscopic injection of botulinum toxin.
    e. robotic-assisted augmentation cystoplasty.
7. The use of antimuscarinics for the treatment of detrusor overactivity in children does NOT result in the following:
    a. Increased bladder capacity
    b. Decreased number of bladder contractions
    c. Decreased number of incontinence episodes
    d. Decreased number of catheterizations

e. Decreased volume to first bladder contraction

8. A possible predictor of poor clinical response to intravesical injection of botulinum toxin injection is preexisting:
   a. low maximum cystometric capacity.
   b. detrusor overactivity.
   c. more than five episodes of incontinence per day.
   d. poor detrusor compliance.
   e. previous botulinum toxin injections.
9. Which of the following increases the risk of developing lower urinary tract stones in children with bladder augmentation?
   a. Use of the ileal segment
   b. Routine bladder irrigation with water or saline
   c. Use of an antimuscarinic
   d. Catheterization of the urethra, rather than an abdominal stoma
   e. A mobile patient
10. The presentation of bladder malignancy in those with a history of bladder augmentation does NOT include:
   a. presenting with advanced disease.
   b. presenting at an older age than is typical for bladder malignancies.
   c. presenting with atypical symptoms such as vague abdominal pain, urosepsis or increased frequency of urinary tract infection (UTI), difficult catheterization, and renal failure.
   d. presenting with a time lag of a minimum of 10 years after a bladder augmentation.
   e. presenting with atypical signs such as new hydronephrosis and bladder wall thickening.
11. Which of the following is the preferred approach to increase bladder capacity in children with neuromuscular dysfunction of the lower urinary tract?
   a. Autoaugmentation
   b. Enteric augmentation with a gastric segment
   c. Tissue engineered bladder substitute
   d. Enteric augmentation with an ileal segment
   e. Enteric augmentation with an ileal-cecal segment

12. Worsening of bladder function after isolated bladder neck procedures including implantation of an artificial urinary sphincter or bladder neck fascial sling is more common in those with:
    a. preexisting detrusor overactivity and poor compliance.
    b. preexisting low maximum cystometric capacity.
    c. surgery in the post-pubertal period.
    d. detrusor sphincter dyssynergia.
    e. history of prior bladder neck outlet procedures.
13. Which of the following is NOT an acceptable method of managing high-grade vesicoureteral reflux in children with neuromuscular dysfunction of the lower urinary tract?
    a. CIC
    b. Antimuscarinics
    c. Antibiotic prophylaxis
    d. Ureteral re-implantation
    e. Bladder emptying by the Credé maneuver
14. Prenatal surgery for children with myelomeningocele compared to standard post-natal closure has been noted to result in:
    a. an increased risk of fetal death and need for cerebrospinal fluid shunting.
    b. worsening of mental development and motor function at 30 months.
    c. fewer pregnancy complications.
    d. a lower risk of preterm labor.
    e. no improvement in bladder function.
15. Indications for the initiation of CIC in the newborn with myelomeningocele do NOT include:
    a. post-void residual urine measurement of 3 mL after the Credé maneuver.
    b. post-void residual urine measurement of 10 mL after spontaneous voiding.
    c. the presence of detrusor sphincter dyssynergia on urodynamic studies.
    d. the presence of hydronephrosis and high-grade vesicoureteral reflux with poor bladder emptying.

e. poor bladder compliance with bladder filling pressures greater than 40 cm of water.

16. The highest risk for the development of urinary tract deterioration in children with myelodysplasia is in those with initial urodynamic findings of:
    a. detrusor sphincter synergy.
    b. detrusor sphincter dyssynergy.
    c. complete denervation.
    d. low maximum cystometric capacity.
    e. detrusor overactivity.
17. The gold standard for measuring renal function in children with spina bifida is:
    a. serum creatinine.
    b. glomerular filtration rate as estimated by the Schwarz formula.
    c. glomerular filtration rate as estimated by the Schwartz formula.
    d. serum cystatin C.
    e. nuclear renography.
18. Sexual function and satisfaction in men with myelomeningocele is better with the following condition:
    a. Living with their parents
    b. Severe incontinence
    c. A sacral level lesion of the neural tube defect
    d. A thoracic level lesion of the neural tube defect
    e. The presence of hydrocephalus
19. The most common finding associated with an occult neural tube defect is:
    a. a cutaneous abnormality overlying the lower spine.
    b. high-arched feet.
    c. claw or hammer toes.
    d. abnormal gait.
    e. absent perineal sensation.
20. In a 1-year-old child, definitive diagnosis of an occult neural tube defect can be made by:
    a. spinal ultrasound.

b. urodynamic studies demonstrating findings consistent with neurogenic bladder dysfunction.
c. magnetic resonance imaging of the spine.
d. documentation of resolution of abnormal urodynamic findings after a detethering procedure.
e. abnormal electromyography of the external urinary sphincter.

21. Which of the following is UNLIKELY to be noted in a child with neuromuscular dysfunction of the lower urinary tract secondary to sacral agenesis?
    a. Urinary incontinence
    b. A maternal history of diabetes mellitus or gestational diabetes
    c. Flattened buttocks and a short, low gluteal cleft
    d. Absent perineal sensation
    e. Vesicoureteral reflux and recurrent urinary tract infections
22. Urodynamic studies of children with an anorectal malformation should be performed in all of the following circumstances EXCEPT:
    a. a bony malformation of the spine or a spinal cord defect.
    b. hydronephrosis.
    c. vesicoureteral reflux.
    d. urinary or fecal incontinence.
    e. a low insertion of the fistulous site.
23. The following statement concerning bladder function in children with cerebral palsy is TRUE:
    a. They achieve nighttime urinary continence first.
    b. They achieve urinary continence at the same age as their age-adjusted peers.
    c. Lower urinary tract symptoms are more common in younger children.
    d. The most common urinary tract symptom is monosymptomatic nocturnal enuresis.
    e. Clinical symptoms of recurrent UTI and detrusor sphincter dyssynergia (retention, interrupted stream, and hesitancy) are associated with upper urinary tract deterioration.
24. The most common presenting urinary symptom in children with transverse myelitis is:

a. urinary incontinence.
b. urinary tract infection.
c. urinary retention.
d. urinary frequency.
e. urinary urgency.

AnswerS

1. **b. Cerebral palsy.** Cerebral palsy is the only acquired disease process on the list. The remainder are congenital.
2. **c. Preservation of renal function.** Although the other listed goals have important clinical and social implications, preservation of renal function is the primary goal of treatment. Achievement of urinary/fecal continence, sexual function, and avoidance of urinary tract infection are secondary goals of treatment.
3. **b. Newborn to toddler AND adolescent to adult.** The correct combination of growth periods in which the rate of somatic growth is highest is in the newborn to toddler and adolescent to adult age group. This recommendation is due to the fact that the highest rate of spinal cord tethering, and thus a change in bladder/bowel function, is during periods of highest somatic growth.
4. **e. Improved continence.** Urinary tract infections, hydronephrosis, worsening continence, and a change in lower extremity function are more likely to indicate a significant change in bladder function than improved continence.
5. **d. Detrusor sphincter dyssynergia.** Detrusor sphincter dyssynergia is the only entity listed that may compromise bladder emptying. All other listed options would facilitate or encourage early bladder emptying.
6. **b. Antimuscarinics and clean intermittent catheterization (CIC).** The beneficial effects of early initiation of antimuscarinics and CIC are well established. The other listed options would be considered to be more invasive or less established methods of intervention.

7. **e. Decreased volume to first bladder contraction.** Antimuscarinics in children result in increased bladder capacity, fewer bladder contractions, less incontinence, and catheterizations. The question is worded negatively; (e) is correct as antimuscarinics would be expected to increase the volume to first bladder contraction, rather than to decrease it.
8. **d. Poor detrusor compliance.** Kask et al. (2013) demonstrated that preexisting poor detrusor compliance predicts a poor clinical response to intravesical injection of botulinum toxin. One could postulate that the histologic changes that are present after the development of poor detrusor compliance are not amenable to the effects of botulinum toxin.
9. **a. Use of the ileal segment.** The use of an ileal segment has been shown to increase the risk of lower urinary tract stones. Catheterization through the urethra, rather than via an abdominal stoma, is associated with fewer lower urinary tract stones, presumably because of better drainage. A mobile patient and routine bladder irrigation decrease the risk of stones. The use of an antimuscarinic should not affect stone formation.
10. **b. Presenting at an older age than is typical for bladder malignancies.** Those with bladder augmentation present at a younger age, with atypical symptoms and advanced disease, usually with a minimum 10-year lag time between augmentation and presentation of bladder malignancy.
11. **d. Enteric augmentation with an ileal segment.** Lack of improvement in urodynamic and clinical symptoms, along with a high failure rate, makes autoaugmentation an undesirable option. Gastric segments are associated with more symptomatic side effects (such as hematuria dysuria syndrome) and metabolic complications than ileal segments. It is recommended to avoid the ileal-cecal segment in children with neural tube defects because it may aggravate bowel dysfunction. Phase II studies of tissue-engineered bladder revealed no improvement in urodynamic parameters and serious adverse events. Thus, the ileal segment is the most desirable for enteric augmentation.

12. **a. Preexisting detrusor overactivity and poor compliance.** Isolated bladder neck procedures such as artificial urinary sphincter and bladder neck repairs have been noted to lead to worsening of bladder function in up to 30% of patients, especially those with preexisting detrusor overactivity and poor compliance. The remaining answers have not been associated with worsening of bladder function after isolated bladder neck procedures.
13. **e. Bladder emptying by the Credé maneuver.** Children with a reactive external urinary sphincter will have a reflex response of increased external urethral tone in response to the Credé maneuver that can aggravate reflux and thus is contraindicated in this group. The other listed options are all reasonable methods of managing vesicoureteral reflux in those with neuromuscular dysfunction of the lower urinary tract.
14. **e. No improvement in bladder function.** Prenatal surgery for children with myelomeningocele compared to standard post-natal closure has been noted to decrease the need for cerebrospinal fluid shunting, improve neuromotor function, and increase pregnancy complications and the risk for preterm labor. The correct answer is there is no improvement in lower urinary tract function with prenatal versus postnatal closure.
15. **a. Post-void residual urine measurement of 3 mL after the Credé maneuver.** A post-void residual urine measurement on 3 mL after the Credé maneuver is within normal limits and does not require the initiation of CIC. All other possible answers are indications for the initiation of CIC.
16. **b. Detrusor sphincter dyssynergy.** Within the first 3 years of life, over 70% of children with detrusor sphincter dyssynergy will have urinary tract deterioration, whereas less than one-quarter of those with synergy or complete denervation will have deterioration. Low maximum cystometric capacity and detrusor overactivity have not been associated with upper tract deterioration.
17. **e. Nuclear renography.** Although all possible answers are means of measuring renal function in children with myelodysplasia, nuclear renography is considered to be the gold standard.

18. **c. A sacral level lesion of the neural tube defect.** Lesions above the sacral spinal cord, hydrocephalus, incontinence, and living with your parents are associated with less sexual function and satisfaction compared to men with a sacral level lesion.
19. **a. A cutaneous abnormality overlying the lower spine.** A cutaneous abnormality overlying the spine such as a skin dimple, tuft of hair, skin tag, lipoma, vascular malformation, or asymmetric gluteal cleft is noted in 90% of those with occult neural tube defect. The other listed answers are also noted as part of this entity but their occurrence is much less frequent.
20. **c. Magnetic resonance imaging of the spine.** Spinal ultrasound may be utilized prior to ossification of vertebral bones (3 months of age). After this time point, magnetic resonance imaging of the spine is required for definitive diagnosis of an occult neural tube defect.
21. **d. Absent perineal sensation.** Most children with sacral agenesis have preserved perineal sensation, thus they are UNLIKELY to have absent perineal sensation. The other listed answers are all potential presenting features of a child with sacral agenesis.
22. **e. A low insertion of the fistulous site.** Urodynamic evaluation in those with anorectal malformation is indicated if there is suggestion of possible spinal cord tethering (bony malformation of the spine or spinal cord defect) or signs and/or symptoms that might indicate a neurogenic defect such as hydronephrosis, vesicoureteral reflux, or urinary/fecal incontinence. Although spinal cord defects may be present in up to a third of those with a low insertion of the fistula site, a low insertion site itself is not an indication for urodynamic study.
23. **e. Clinical symptoms of recurrent UTI and detrusor sphincter dyssynergia (retention, interrupted stream, and hesitancy) are associated with upper urinary tract deterioration.** Children with cerebral palsy usually achieve daytime dryness first, at an age later than their age-adjusted peers. Lower urinary tract symptoms become more prevalent in children with cerebral palsy as they age. The most common symptom in these children is incontinence. Clinical symptoms of recurrent UTI and detrusor sphincter

dyssynergia are associated with upper urinary tract deterioration and may warrant investigation with renal/bladder ultrasound and urodynamic studies.

24. **c.** The most common presenting urinary symptom of transverse myelitis is urinary retention. Ninety-five percent of children in the acute phase of the disease will have urinary retention.

Chapter Review

1. Renal ultrasonography and measurement of residual urine are performed as early as possible after birth. Initial evaluation can be compared with findings on subsequent assessments, so that early signs of deterioration of urinary tract function and drainage, or of progressive neurologic denervation, can be detected.
2. Infants at risk for urinary tract deterioration as a result of a poorly compliant or overactive detrusor or outflow obstruction from detrusor-sphincter dyssynergia need to be identified. This determines the need to initiate prophylactic measures before any deterioration in upper urinary tract architecture and function take place.
3. Three categories of lower urinary tract (LUT) dynamics may be detected: synergic (26%), dyssynergic with and without poor detrusor compliance (37%), and complete denervation (36%).
4. Early intervention with CIC and antimuscarinics improves urodynamic parameters and decreases the rate of UTI, vesicoureteral reflux (VUR), upper urinary tract deterioration, and the incidence of end-stage renal disease.
5. Cystatin C is superior to serum creatinine in monitoring renal function in children with neural tube defects. Renography is the gold standard for estimating glomerular filtration rate (GFR).
6. Determinants of risk of renal dysfunction in spina bifida include: detrusor sphincter dyssynergia, high detrusor pressures, detrusor overactivity, febrile UTIs, and vesicoureteral reflux. These entities should be actively managed to minimize risk of renal functional decline.

7. The primary goal of management is preservation of renal function. Secondary goals of management include urinary/fecal continence, avoidance of urinary tract infection, and facilitation of sexual function and fertility.
8. Preservation of renal function is achieved by maintaining low bladder filling pressures and active management of vesicoureteral reflux and avoidance of urinary tract infection.
9. The International Children's Continence Society recommendations for follow-up are based on developmental stages and the relative risk for secondary spinal cord tethering.
10. The CDC-endorsed multicenter collaborative has published a protocol for the management of patients with neurogenic bladder ages from birth to 5 years.
11. Minimally invasive approaches should precede the use of more invasive therapies to address bladder failure to store urine efficiently or empty effectively. Antimuscarinic therapy and CIC are the mainstay of therapy.
12. Bladder filling pressures should be maintained below 30 cm H_2O as much as possible to prevent upper urinary tract deterioration.
13. Intravesical injection of botulinum toxin is well tolerated and avoids the need for more invasive treatment options in a high proportion of children with minimal side effects.
14. Enterocystoplasty is an effective option to prevent upper urinary tract deterioration but comes with a high risk of complications.
15. Patients undergoing isolated bladder neck procedures (artificial urinary sphincter and bladder neck sling) may develop detrusor overactivity and poor compliance that leads to upper urinary tract deterioration, mandating close follow-up.
16. Low-grade reflux in those who are emptying well without outlet resistance can be managed with antibiotic prophylaxis alone. High-grade reflux or those who do not empty well also require CIC.
17. Those with poor compliance with or without hydronephrosis should be started on antimuscarinics.
18. The Credé maneuver is contraindicated in patients with reflux who have a reactive external urethral sphincter.

19. Indications for antireflux surgery are similar to those for children without neurogenic bladder. Success rates are similar if effective bladder emptying is addressed.
20. Children with an occult neural tube defect often present after age 3 or 4 years with incontinence or constipation. They may present with new urologic symptoms after a growth spurt related to tethering of the spinal cord. Most have a cutaneous abnormality overlying the spine.
21. Radiologic examination of the spine, MRI, and renal US are indicated in the initial evaluation. A spinal US may be used as a screening investigation prior to MRI in those less than 3 months of age.
22. Urodynamic studies should be done before and after spinal cord detethering.
23. Intervention with spinal cord detethering early in life is associated with better outcomes than when done at a later age.
24. Spinal cord abnormalities including a tethered spinal cord, thickened or fatty filum terminale, and a lipoma occur in 18% to 50% of patients with an anorectal malformation.
25. The incidence of LUTS in children with cerebral palsy is 16% to 94%, and the most common symptom is urinary incontinence.
26. In patients with spinal cord injury, there is an acute phase of spinal shock wherein the bladder is often acontractile and the urethral sphincter nonreactive; with resolution of spinal shock, an overactive detrusor and bladder-sphincter dyssynergy most commonly develop.
27. Patients with a cervical or thoracic spine lesion above T6 are likely to exhibit autonomic dysreflexia during bladder filling and detrusor contractions.